

Assignment #1 - CPCS 222 (Spring 2026) (10 points)
(Boy's campus)

Due: (30/04/2026)

Student Name : _____ Student ID: _____

Section : _____

1. Construct a truth table for $(a \oplus \neg b) \leftrightarrow (\neg a \rightarrow c)$

[1.5 points]

a	b	c	$\neg a$	$\neg b$			
T							
T							
T							
T							
F							
F							
F							
F							

2. Translate the English statement into propositional logic - *"If the bus and*

Uber are parked at the bus station, then passengers can travel to Riyadh"

[0.5 points]

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3. Write the converse and inverse of the propositional logic obtained from question 2. Also, write these in English.

[1 point]

Converse	
Inverse	

4. Show that $[p \wedge (p \rightarrow q)] \rightarrow q$ is a tautology

[2 points]

Expression	Rule

5. Translate the English statements into predicate logic where the domain of x is all animals in the world.

Hint: Use $f(x)$ – x is an animal in Riyadh Zoo , $j(x)$ – x is a fish and $m(x)$ – x is a lion

A) “Some animals in Riyadh Zoo are fishes or lions”

[0.5 points]

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B) Negate the English statement given in question 5

[0.5 points]

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is a consequence of the premises:

- [illegible]